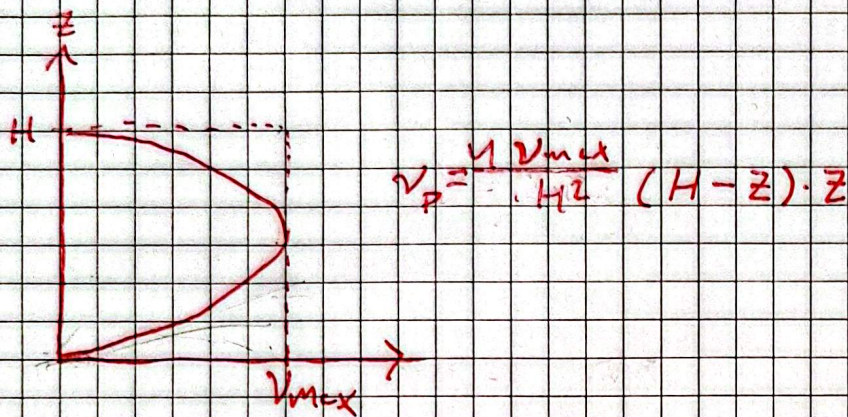


EKman 1.5 Skikt - Parabol ν

D1

Vi varierar viskositeten



där
$$\frac{d}{dz} \left(\nu \frac{du}{dz} \right) = i f (u - u_0) \quad (1)$$

$$u(0) = 0$$

$$\frac{du}{dz} = 0, \quad z = H$$

endigt 1.5 Skiktens modellens argumenterande
utifrån impulsströmmen.

$$(1) \Rightarrow \nu u'' + \nu' u' = i f (u - u_0)$$

Använd $u = u + i v, \quad u_0 \in \mathbb{R}$

$$\Rightarrow \begin{aligned} \nu u'' + \nu' u' &= -f v \\ \nu v'' + \nu' v' &= f (u - u_0) \end{aligned}$$

definiera
$$\begin{cases} y_1 = u \\ y_2 = u' \\ y_3 = v \\ y_4 = v' \end{cases} \quad \begin{aligned} y_1' &= y_2 \\ y_3' &= y_4 \end{aligned}$$

$$\Rightarrow \begin{cases} y_2' = \frac{1}{\nu} (-f y_3 - \nu' y_2) \\ y_4' = \frac{1}{\nu} (f (y_1 - u_0) - \nu' y_4) \end{cases}$$

Men vill beskriva problemet med en dimensionslös variabel

$$f = \frac{H}{H_{\text{ref}}} \sim H \sqrt{\frac{f}{\nu_{\text{max}}}} \quad (2)$$

Detta liser vi genom dimensionslös höjd koordinat

$$\hat{z} = z/H, \quad \hat{z} \in [0, 1] \quad (3)$$

derivatn blir då $d/dz = \frac{1}{H} d/d\hat{z}$
då blir viskositeten

$$\nu_p = \nu \nu_{\text{max}} \hat{z} (1 - \hat{z})$$

$$(1) \Rightarrow \frac{d}{d\hat{z}} \left(\frac{1}{H^2} \nu_p \frac{du}{d\hat{z}} \right) = i f (u - u_0)$$

$$\Rightarrow \frac{d}{d\hat{z}} \left(\nu \hat{z} (1 - \hat{z}) \frac{du}{d\hat{z}} \right) = i \frac{1}{\nu_{\text{max}}} (u - u_0)$$

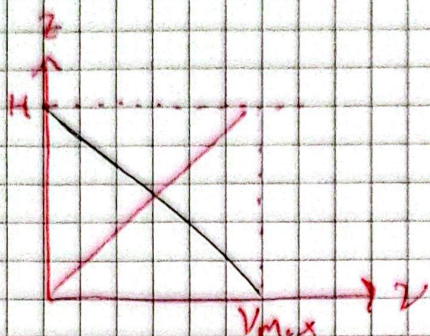
$$\Rightarrow \frac{d}{d\hat{z}} \left(\underbrace{\nu \hat{z} (1 - \hat{z})}_{\hat{\nu}_p} \frac{du}{d\hat{z}} \right) = i f^2 (u - u_0) \quad (5)$$

$$\hat{\nu}_p = \nu_p$$

EKman 1.5 Skut - linjär v

D3

V_i varierar viskositeten linjärt



$$v_{\uparrow} = \frac{V_{\max}}{H} z \quad (1)$$

$$v_{\downarrow} = \frac{V_{\max}}{H} (H - z) \quad (2)$$

der V_i har

$$\frac{d}{dz} \left(\nu \frac{du}{dz} \right) = i \tau (u - u_0) \quad \begin{matrix} u(0) = 0 \\ \frac{du}{dz}|_H = 0 \end{matrix} \quad (3)$$

$$\tau \sim H \sqrt{\frac{\rho}{\nu_{\max}}} \quad (4)$$

Anser dimensionlösningen der

$$\hat{z} = \frac{z}{H}, \quad (5)$$

$$\Rightarrow \frac{d}{dz} = \frac{1}{H} \frac{d}{d\hat{z}} \quad \begin{matrix} \hat{v}_{\uparrow} = V_{\max} \hat{z} \\ \hat{v}_{\downarrow} = V_{\max} (1 - \hat{z}) \end{matrix}$$

Skriver om för (3):

$$v_{\uparrow}: \frac{d}{d\hat{z}} \left(\frac{1}{H^2} \nu \frac{du}{d\hat{z}} \right) = i \tau (u - u_0)$$

$$\Leftrightarrow \frac{d}{d\hat{z}} \left(\frac{1}{H^2} \nu_{\max} \hat{z} \frac{du}{d\hat{z}} \right) = i \tau (u - u_0)$$

$$\Leftrightarrow \frac{d}{d\hat{z}} \left(\underbrace{\hat{z}}_{\hat{v}_{\uparrow}} \frac{du}{d\hat{z}} \right) = i \tau^2 (u - u_0) \quad (6)$$

$$v_{\downarrow}: \dots \Leftrightarrow \frac{d}{d\hat{z}} \left(\underbrace{[1 - \hat{z}]}_{\hat{v}_{\downarrow}} \frac{du}{d\hat{z}} \right) = i \tau^2 (u - u_0) \quad (7)$$